

## TPS System Overview

TP01X-400

TPS R400

3/2007

## Notices and Trademarks

Copyright 2007- by Honeywell International Inc.  
TPS R400 3/2007

While this information is presented in good faith and believed to be accurate, Honeywell disclaims the implied warranties of merchantability and fitness for a particular purpose and makes no express warranties except as may be stated in its written agreement with and for its customers.

In no event is Honeywell liable to anyone for any indirect, special or consequential damages. The information and specifications in this document are subject to change without notice.

Honeywell, PlantScape, Experion, **TotalPlant**, Business.FLEX, SafeView, FSC, and QMR are registered trademarks of Honeywell International Inc.

UserAlert is a trademark of Honeywell International Inc.

Other brand or product names are trademarks of their respective owners.

Honeywell International  
Process Solutions  
2500 West Union Hills Drive  
Phoenix, AZ 85027  
**1-800 223-8947**

## About This Document

This document describes the TPS System, its interfaces to control systems, and its management of networks and communications.

### Release Information

| Document Name               | Document ID | Release Number | Publication Date |
|-----------------------------|-------------|----------------|------------------|
| TPS System Overview - TP01X | TP01X-400   | TPS R400       | 3/2007           |

### Contacts

#### World Wide Web

The following Honeywell web sites may be of interest to Honeywell Process Solutions customers.

| Honeywell Organization | WWW Address (URL)                                                     |
|------------------------|-----------------------------------------------------------------------|
| Corporate              | <a href="http://www.honeywell.com">http://www.honeywell.com</a>       |
| Process Solutions      | <a href="http://www.honeywell.com/ps">http://www.honeywell.com/ps</a> |

## Telephone

Contact us by telephone at the numbers listed below.

| Location                   | Organization                           | Phone                                                                                   |
|----------------------------|----------------------------------------|-----------------------------------------------------------------------------------------|
| United States and Canada   | Honeywell IAC Solution Support Center  | 1-800-822-7673                                                                          |
| Europe                     | Honeywell TAC-EMEA                     | +32-2-728-2704                                                                          |
| Pacific                    | Honeywell Global TAC - Pacific         | 1300-300-4822<br>(toll free within Australia)<br>+61-8-9362-9559<br>(outside Australia) |
| India                      | Honeywell Global TAC - India           | +91-20-2682-2458 / 1600-44-5152                                                         |
| Korea                      | Honeywell Global TAC - Korea           | +82-2-799-6317                                                                          |
| People's Republic of China | Honeywell Global TAC - China           | +86-10-8458-3280 ext. 361                                                               |
| Singapore                  | Honeywell Global TAC - South East Asia | +65-6580-3500                                                                           |
| Taiwan                     | Honeywell Global TAC - Taiwan          | +886-7-323-5900                                                                         |
| Japan                      | Honeywell Global TAC - Japan           | +81-3-5440-1303                                                                         |
| Elsewhere                  | Call your nearest Honeywell office.    |                                                                                         |

## Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

| Symbol                                                                              | Definition                                                                                                |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
|  | <b>REFERENCE - INTERNAL:</b> Identifies an additional source of information within the Honeywell bookset. |

# Contents

|                                                                              |           |
|------------------------------------------------------------------------------|-----------|
| <b>1. SYSTEM INTRODUCTION.....</b>                                           | <b>9</b>  |
| <b>1.1 An Open Automation System .....</b>                                   | <b>9</b>  |
| Introduction .....                                                           | 9         |
| System features .....                                                        | 10        |
| TPS system uses Open technologies .....                                      | 15        |
| Adding Experion to TPS .....                                                 | 16        |
| <b>1.2 TPS Expands the Control Domain.....</b>                               | <b>18</b> |
| Introduction .....                                                           | 18        |
| Expanded control domain benefits .....                                       | 18        |
| <b>2. INTERFACING TO CONTROL SYSTEMS.....</b>                                | <b>19</b> |
| <b>2.1 Introduction .....</b>                                                | <b>19</b> |
| About this section .....                                                     | 19        |
| <b>2.2 Experion Station – TPS.....</b>                                       | <b>20</b> |
| About the Experion Station – TPS.....                                        | 20        |
| <b>2.3 Experion Station – Console Extension .....</b>                        | <b>21</b> |
| About the Experion Station – Console Extension.....                          | 21        |
| <b>2.4 Experion Station – Flex .....</b>                                     | <b>22</b> |
| About the Experion Station – Flex .....                                      | 22        |
| <b>2.5 Global User Station.....</b>                                          | <b>23</b> |
| About the Global User Station.....                                           | 23        |
| <b>2.6 Icon Series Console.....</b>                                          | <b>24</b> |
| About the Icon Series console.....                                           | 24        |
| <b>2.7 Operator Station Displays .....</b>                                   | <b>25</b> |
| Access to control and process information.....                               | 25        |
| Types of displays .....                                                      | 25        |
| <b>2.8 SafeView.....</b>                                                     | <b>26</b> |
| About SafeView.....                                                          | 26        |
| <b>2.9 GUSPro.....</b>                                                       | <b>27</b> |
| Applications for alarm management, event analysis and advanced trending..... | 27        |
| Using GUSPro tools .....                                                     | 27        |
| Developed in conformance with ASM practices .....                            | 27        |
| ProMenu.....                                                                 | 28        |

|                                                     |           |
|-----------------------------------------------------|-----------|
| UserAlert™ .....                                    | 29        |
| Alarm Configuration Manager.....                    | 30        |
| Event Analyst and ProTrend .....                    | 31        |
| <b>2.10 Experion Server – TPS.....</b>              | <b>32</b> |
| About the Experion Server – TPS .....               | 32        |
| <b>3. CONTROLLING THE PROCESS .....</b>             | <b>33</b> |
| <b>3.1 Honeywell Process Control Products .....</b> | <b>33</b> |
| Helping you to control your world .....             | 33        |
| <b>3.2 High-Performance Process Manager.....</b>    | <b>34</b> |
| About the High-Performance Process Manager .....    | 34        |
| HPM architecture.....                               | 34        |
| <b>3.3 Fail Safe Controller.....</b>                | <b>35</b> |
| Help safeguard your process operations.....         | 35        |
| <b>3.4 Honeywell Field Instruments .....</b>        | <b>36</b> |
| Accurate, reliable measurement.....                 | 36        |
| Supporting FOUNDATION Fieldbus.....                 | 36        |
| <b>3.5 TPS Field I/O.....</b>                       | <b>38</b> |
| Field devices for extreme environments.....         | 38        |
| <b>3.6 TPS System Tools .....</b>                   | <b>39</b> |
| TPS Builder .....                                   | 40        |
| HMIWeb Display Builder .....                        | 42        |
| HMIWeb Faceplates for TPN points.....               | 43        |
| <b>3.7 Accessing Experion Controllers .....</b>     | <b>44</b> |
| Accessing Experion controllers from TPS .....       | 44        |
| Direct Control Integration .....                    | 45        |
| LCN-connected ACE .....                             | 45        |
| <b>4. MANAGING SYSTEMS AND NETWORKS.....</b>        | <b>47</b> |
| <b>4.1 System Management .....</b>                  | <b>47</b> |
| About System Management .....                       | 47        |
| What does System Management provide?.....           | 47        |
| Experion configuration tools .....                  | 49        |
| <b>4.2 System Management Operations .....</b>       | <b>50</b> |
| Introduction to System Management.....              | 50        |
| What is the System Management display? .....        | 50        |

---

|            |                                                                        |           |
|------------|------------------------------------------------------------------------|-----------|
| <b>4.3</b> | <b>TPS Domains and Consoles .....</b>                                  | <b>51</b> |
|            | Using Windows 2003 system configuration tools .....                    | 52        |
| <b>5.</b>  | <b>SUPPORTING CONTROL COMMUNICATIONS .....</b>                         | <b>53</b> |
| <b>5.1</b> | <b>Open System Technologies and Products .....</b>                     | <b>53</b> |
|            | Introduction .....                                                     | 53        |
|            | Open System technologies .....                                         | 53        |
|            | TPS system communications products and technologies .....              | 53        |
| <b>5.2</b> | <b>COM/DCOM and OPC .....</b>                                          | <b>54</b> |
|            | About system support .....                                             | 54        |
|            | COM/DCOM .....                                                         | 54        |
|            | DCOM works with HCI/OPC .....                                          | 54        |
| <b>5.3</b> | <b>HCI/OPC .....</b>                                                   | <b>55</b> |
|            | About HCI/OPC .....                                                    | 55        |
| <b>5.4</b> | <b>Fault Tolerant Ethernet .....</b>                                   | <b>55</b> |
|            | About Fault Tolerant Ethernet .....                                    | 55        |
|            | Using FTE with new and existing systems .....                          | 55        |
|            | How does FTE work? .....                                               | 56        |
| <b>5.5</b> | <b>Redirection Manager .....</b>                                       | <b>57</b> |
|            | About Redirection Manager .....                                        | 57        |
| <b>5.6</b> | <b>Distributed System Architecture .....</b>                           | <b>58</b> |
|            | About Distributed System Architecture .....                            | 58        |
| <b>6.</b>  | <b>INTEGRATING WITH CONTROL AND BUSINESS NEEDS .....</b>               | <b>59</b> |
| <b>6.1</b> | <b>Application Processing Platform .....</b>                           | <b>59</b> |
|            | About the Application Processing Platform .....                        | 59        |
| <b>6.2</b> | <b>Control and Business Process Integration .....</b>                  | <b>60</b> |
|            | PHD software integrates business and control information. ....         | 60        |
| <b>6.3</b> | <b>Business.FLEX: Improved Decision Making and Profitability .....</b> | <b>61</b> |
| <b>6.4</b> | <b>Services to Protect Your Technology Investment .....</b>            | <b>62</b> |
|            | Solution Enhancement Support Services .....                            | 62        |
| <b>6.5</b> | <b>Consistent Evolution .....</b>                                      | <b>62</b> |

## Tables

|                                   |    |
|-----------------------------------|----|
| Table 1 TPS System Products ..... | 10 |
|-----------------------------------|----|

## Figures

|                                                                                 |    |
|---------------------------------------------------------------------------------|----|
| Figure 1 TPS integrates business, information, and control systems .....        | 9  |
| Figure 2 TPS is an open system .....                                            | 15 |
| Figure 3 Experion components on TPS .....                                       | 17 |
| Figure 4 TPS unifies business and process control domains .....                 | 18 |
| Figure 5 Honeywell Icon Series console .....                                    | 24 |
| Figure 6 SafeView helps you to view critical plant operations .....             | 26 |
| Figure 7 Honeywell controllers and field instruments integration .....          | 33 |
| Figure 8 Typical Honeywell Fieldbus configuration .....                         | 37 |
| Figure 9 Field I/O configuration .....                                          | 38 |
| Figure 10 TPS Builder helps you to build and configure control strategies ..... | 39 |
| Figure 11 TPS Builder tools for control strategy display .....                  | 41 |
| Figure 12 HMIWeb Display Builder gallery .....                                  | 42 |
| Figure 13 TPN Point in HMIWeb faceplate .....                                   | 43 |
| Figure 14 Role of system management .....                                       | 47 |
| Figure 15 Experion Configuration Studio display .....                           | 49 |
| Figure 16 Typical System Management display .....                               | 50 |
| Figure 17 TPS Domains in a Windows domain .....                                 | 51 |
| Figure 18 Architecture of HCI/OPC communication infrastructure .....            | 53 |
| Figure 19 How Redirection Manager works .....                                   | 57 |
| Figure 20 Geographically distributed system .....                               | 58 |
| Figure 21 Application Processing Platform (APP) .....                           | 59 |
| Figure 22 Business.FLEX supports collaborative manufacturing .....              | 61 |

# 1. System Introduction

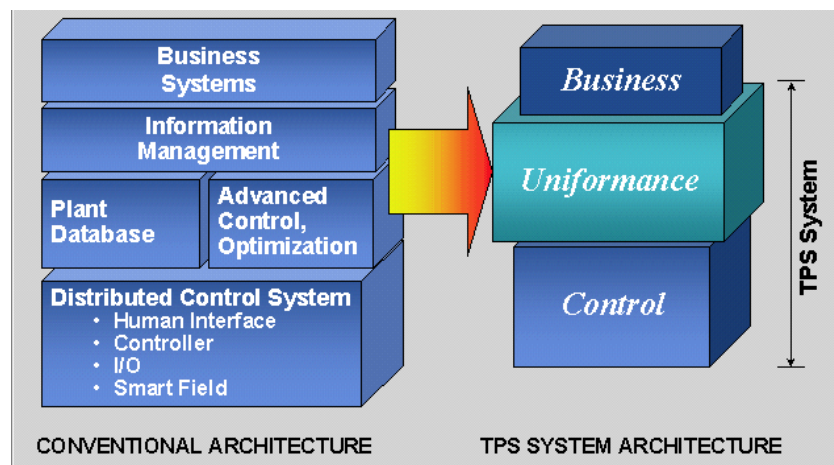
## 1.1 An Open Automation System

### Introduction

The Honeywell TPS System is a Windows-based open plant automation system. It combines robust, secure distributed control capabilities, advanced applications capabilities (such as multivariable control, batch control, optimization, plant-wide history, and information management) into one unified system. This combination of capabilities helps you respond to changes in the marketplace and not compromise the safety and the integrity of your process.

The Honeywell **TotalPlant®** Solution (TPS) platform integrates business systems—from the acquisition of raw materials to global supply chain management and ERP applications—with manufacturing control systems and process control systems. This integration reduces complexity, simplifies management, and improves performance. TPS helps translate the vast collection of data from your systems into reports, trends, and other useful information, which helps you make better-informed decisions. TPS runs on Windows workstations and is designed from the ground up to make use of Windows features and flexibility. The TPS architecture provides flexibility for growth—it supports as many as 94 nodes per system. Designed to meet the needs of the largest systems, the TPS system is also scalable to relatively small systems.

**Figure 1 TPS integrates business, information, and control systems**



## 1. System Introduction

### 1.1. An Open Automation System

---

#### System features

Honeywell provides a wide range of products that help you control your world. These products help you to perform a variety of tasks, including:

- Controlling your process
- Interfacing to control systems
- Supporting control communications
- Managing your enterprise systems and networks
- Integrating with advanced control and business needs.

Table 1 shows TPS system products, grouped by the types of tasks they perform.

**Table 1 TPS System Products**

| Interface to Control Systems |                                                                                                                                                                                                                            |                                              |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>Experion® Stations</b>    | An Experion Station – TPS (ES-T) provides a common operator interface for both TPS and Experion. Directly connects to the LCN through the LCNP board and connects to the Experion controllers through the Experion Server. | See Section <a href="#">2.2</a> for details. |
|                              | Experion Station – Console Extension (ES-CE) connects to the LCN through the Experion Station - TPS. This station supports remote Native Window and remote GUS Displays.                                                   | See Section <a href="#">2.3</a> for details. |
|                              | Experion Station – Flex (ES-F) connects to LCN and Experion controllers through the Experion Server – TPS (ESVT). This station supports remote Native Window.                                                              | See Section <a href="#">2.4</a> for details. |

| Interface to Control Systems                                   |                                                                                                                                                                                                                                                                                                                                                                                                  |                                               |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>Experion Server – TPS (ESVT)</b>                            | Experion Server – TPS directly connects to the LCN through the LCNP board. Provides host and repository services for Experion stations. Integrates with other Experion servers through Distributed System Architecture (DSA).                                                                                                                                                                    | See Section <a href="#">2.10</a> for details. |
| <b>Global User Station</b>                                     | The Global User Station (GUS) is an interface to TPS systems.                                                                                                                                                                                                                                                                                                                                    | See Section <a href="#">2.5</a> for details.  |
| <b>Honeywell Icon Series™ Console</b>                          | Honeywell Icon Series Console is an ergonomically designed console for special operating environments.                                                                                                                                                                                                                                                                                           | See Section <a href="#">2.6</a> for details.  |
| <b>Operator Station Displays</b>                               | Standard operating displays include plant process displays, TPS system displays, and system function displays.                                                                                                                                                                                                                                                                                   | See Section <a href="#">2.7</a> for details.  |
| <b>SafeView®</b>                                               | Honeywell's optional Windows manager application safeguards your plant, personnel and assets by ensuring that critical operating displays are never obstructed by other displays or applications.                                                                                                                                                                                                | See Section <a href="#">2.8</a> for details.  |
| <b>GUSPro™</b>                                                 | GUSPro's integrated operating environment provides powerful tools that help protect profitable operational performance at your plant.                                                                                                                                                                                                                                                            | See Section <a href="#">2.9</a> for details.  |
| <b>A Suite of Advanced Control Applications</b>                | Improve plant production and financial performance.                                                                                                                                                                                                                                                                                                                                              | See Section <a href="#">6</a> for details.    |
| <b>Global Positioning Satellite (GPS) Time Synchronization</b> | GPS Time Synchronization matches LCN time to UTC (Universal Time Coordinated) – the international standard for broadcast used by the array of Global Positioning Satellites. Time synchronization eliminates costs associated with manual checks and adjustments to the TPS system's TPN clock, reconciles time errors in historical data, and manually synchronizes time across different LCNs. |                                               |

## 1. System Introduction

### 1.1. An Open Automation System

---

| Interface to Control Systems                                                                 |                                                                                                                                                                                       |                                              |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>High-Performance Process Manager (HPM)</b>                                                | Contains a set of field-proven algorithms and tools for implementing integrated control strategies.                                                                                   | See Section <a href="#">3.2</a> for details. |
| <b>Fail Safe Controller (FSC®)</b>                                                           | Enhances the safety of your process, plant and personnel. Provides flexibility in designing, configuring, monitoring, controlling and safeguarding your safety-critical applications. | See Section <a href="#">3.3</a> for details. |
| <b>Honeywell Field Instruments</b>                                                           | Integrated instrumentation solutions include products for temperature, pressure, level, flow, and specialty measurements.                                                             | See Section <a href="#">3.4</a> for details. |
| <b>FOUNDATION Fieldbus support</b>                                                           | Industry standard compliance with FOUNDATION Fieldbus brings faster communications from the process to the field                                                                      | See Section <a href="#">3.4</a> for details  |
| <b>TPS Field I/O</b>                                                                         | Field devices designed for extreme environments.                                                                                                                                      | See Section <a href="#">3.5</a> for details  |
| <b>TPS Builder</b>                                                                           | A graphical engineering application for building control strategies and configuring process control data on a TPS system.                                                             | See Section <a href="#">3.6</a> for details  |
| <b>HMIWeb Display Builder</b>                                                                | Display builder based on Web technology.                                                                                                                                              | See Section <a href="#">3.6</a> for details  |
| <b>Plant-wide information management</b><br><b>System Event filter Configuration Utility</b> | Provides detailed analysis for improving process operation.<br><br>Helps you sort and prioritize system events, and displays the events on your screen.                               | See Section <a href="#">4</a> for details.   |
| <b>An open environment</b>                                                                   | Helps you develop advanced applications and manage your plant by making real-time control data available in open, third party applications.                                           | See Section <a href="#">5.1</a> for details. |
| <b>A secure environment</b>                                                                  | Helps safeguard your process operations.                                                                                                                                              | See Section <a href="#">4</a> for details.   |

| Support Microsoft Windows Technologies                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Honeywell uses Microsoft technologies to ensure your TPS system continues to support new Windows platforms as they are released.</p> <p>TPS takes full advantage of Microsoft Windows multi-tasking features to prioritize execution of simultaneously running applications and deliver a high level of responsiveness.</p> <p>See Section 4 for details.</p> <p>Also refer to Microsoft Windows 2003/XP documentation.</p> | <p>Microsoft Windows technologies include:</p> <p><b>IP Multicast</b>—A bandwidth-conserving technology that reduces traffic by delivering a single stream of information simultaneously to potentially thousands of recipients.</p> <p><b>Multicast Group</b>—A group of receivers of a particular data stream. A multicast group has no physical or geographical boundaries. If Hosts are to receive data that flows to a particular group, they must join the group using Internet Group Management Protocol (IGMP).</p> <p><b>IP Multicast Address</b>—Specifies an arbitrary group of IP hosts that have joined a group and want to receive traffic sent to this group.</p> <p><b>Windows Management Instrumentation (WMI)</b>—Provides clients with database-like query capabilities and event notification when a new instance of an object is created, deleted or modified.</p> <p><b>Web-Based Enterprise Management (WBEM)</b>—Provides uniform access to management information. WMI is the Microsoft implementation of this technology.</p> <p><b>Microsoft Management Console (MMC)</b>—An extensible presentation service for system management applications that provides a common integration environment for Microsoft and Honeywell snap-ins.</p> <p><b>System Status Display</b>—Helps you configure and monitor managed nodes and their HCI-managed components.</p> <p><b>System Event Filter Configuration Utility</b>—Helps you create System Event Filter Tables that contain system events you want to monitor.</p> <p><b>Microsoft Installer (MSI) Technology</b>—Uses InstallShield-created MSI packages to install TPS packages.</p> <p><b>Microsoft Web Technology</b>—Uses web display technology as the basis for HMIWeb Display Builder.</p> <p><b>COM, DCOM, and OPC</b> (Component Object Modeling, Distributed COM, and OLE for Process Control)—Industry standards for open communications that provide secure communication between major TPS components distributed across TPS nodes. See Section 5.1 for details.</p> |

## 1. System Introduction

### 1.1. An Open Automation System

---

| Support Honeywell Open System Technologies |                                                                                                                                      |                                              |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>HCI/OPC</b>                             | Honeywell Communications Interface (HCI) for OPC, the industry-standard open communications protocol for process control.            | See Section <a href="#">5.3</a> for details. |
| <b>Fault-Tolerant Ethernet (FTE)</b>       | Fault-Tolerant Ethernet helps increase system reliability, availability and plant productivity.                                      | See Section <a href="#">5.4</a> for details. |
| <b>Redirection Manager (RDM)</b>           | The optional Redirection Manager software enables clients to transparently switch to a secondary server if the primary server fails. | See Section <a href="#">5.5</a> for details. |

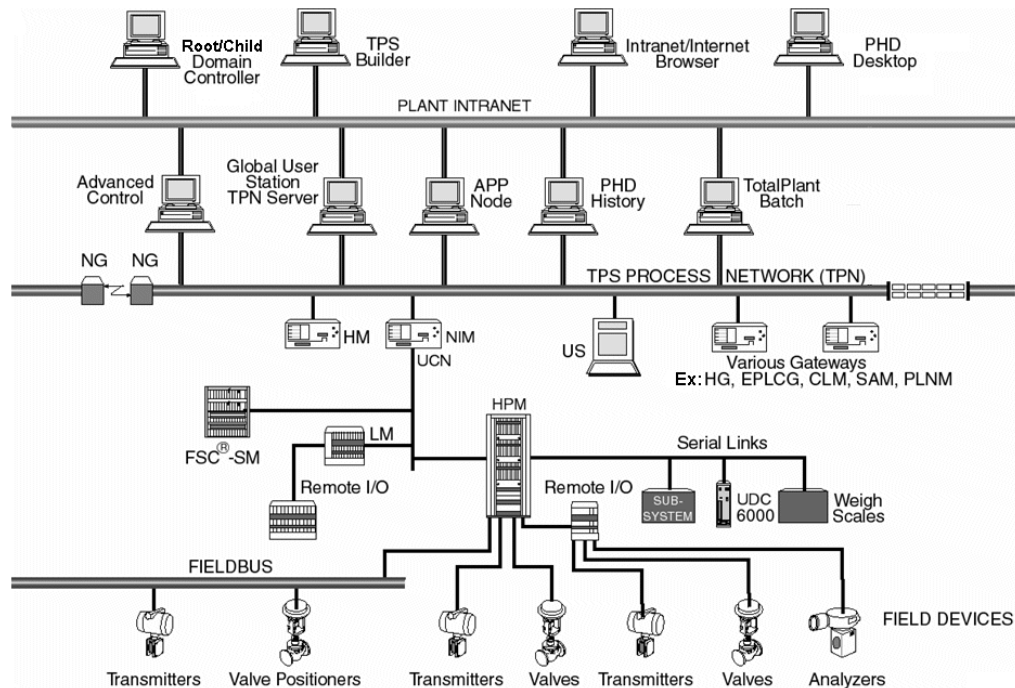
| Integrate Advanced Control and Business Needs |                                                                                                                                                  |                                              |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>APP</b>                                    | The TPS system Advanced Processing Platform (APP) provides advanced control and optimization.                                                    | See Section <a href="#">6.1</a> for details  |
| <b>PHD</b>                                    | The Process History Database (PHD) software establishes an integrated architecture for plant-wide data collection, storage, and management.      | See Section <a href="#">6.2</a> for details. |
| <b>Business FLEX®</b>                         | The optional Business.FLEX software converts business objectives into operational targets and returns validated production data to your desktop. | See Section <a href="#">6.3</a> for details. |
| <b>Services</b>                               | Honeywell offers a wide range of services to protect your technology investment.                                                                 | See Section <a href="#">6.4</a> for details. |
| <b>Consistent Evolution</b>                   | Honeywell ensures that the Honeywell equipment you purchase integrates and migrates easily with future Honeywell products.                       | See Section <a href="#">6.5</a> for details. |

**TPS system uses Open technologies**

TPS provides a platform for open technologies, and includes Object Linking and Embedding (OLE) for Process Control (OPC), FOUNDATION Fieldbus, and Microsoft Windows 2003/XP.

Open standards can play an important part in your automation strategy, because they offer the capability to use leading-edge technology and advanced applications. At the same time, open systems technologies promise greater interoperability and lower life cycle costs.

**Figure 2 TPS is an open system**



## **1. System Introduction**

### **1.1. An Open Automation System**

---

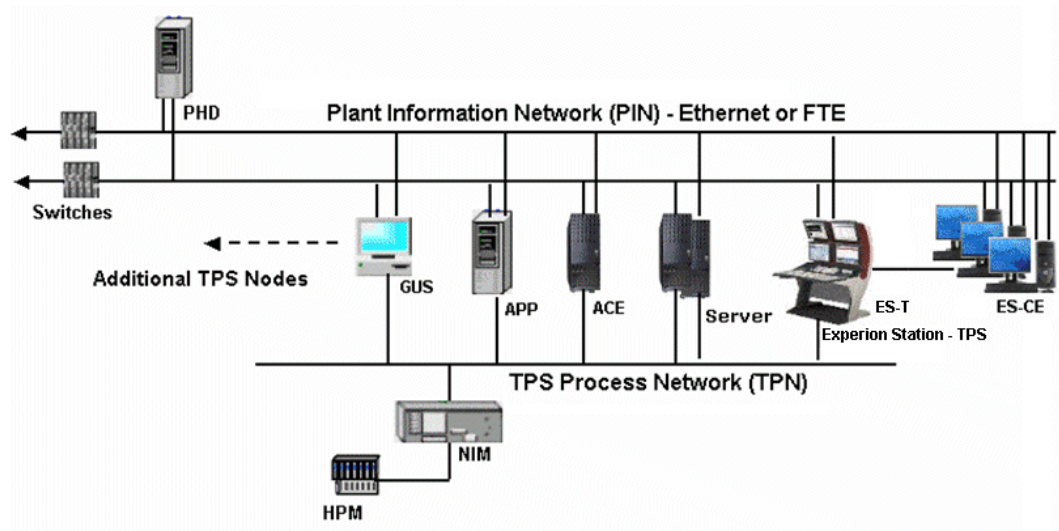
#### **Adding Experion to TPS**

TPS can integrate with or migrate to Experion. One Experion Server (or one redundant pair) is required for each TPN (also referred to as the LCN) that is to be integrated or migrated. One or more TPN-connected Experion Stations (Experion Station – TPS) can be added to each TPN that includes a TPN-connected Experion Server (Experion Server – TPS)). The Experion Station – Flex (ES-F) can also be added as a client to the Server, but it does not connect directly to the TPN (through an LCNP board) as does the ES-T and ESVT. For more information on these Experion components, refer to the following sections in this document:

- Section 2.2 Experion Station – TPS
- Section 2.3 Experion Station – Console Extension
- Section 2.4 About the Experion Station – Flex
- Section 2.10 Experion Server – TPS
- Section 3.7 Accessing Experion Controllers

The following figure shows the addition of Experion nodes such as an ACE, Server, ES-T, and ES-CE to a TPS System.

**Figure 3 Experion components on TPS**



## 1. System Introduction

### 1.2. TPS Expands the Control Domain

---

## 1.2 TPS Expands the Control Domain

### Introduction

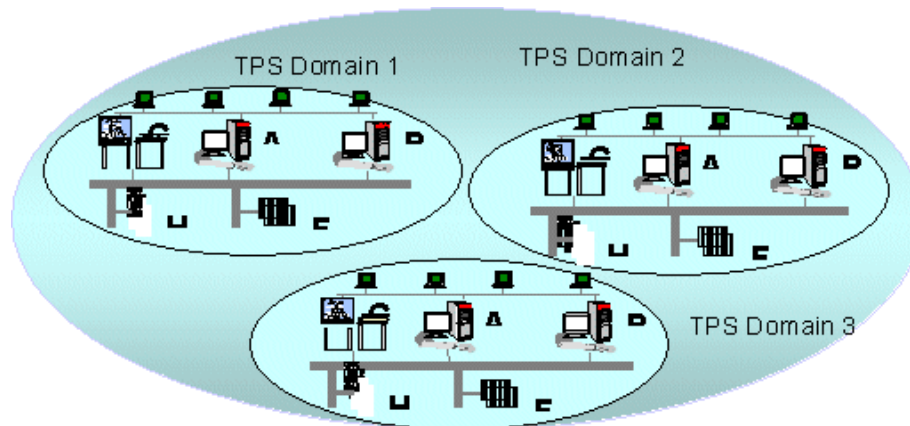
TPS makes it simple to leverage business and control information to attain corporate objectives and achieve real-time plant operations management. Integrating your business and control domains provides transparent integration throughout the enterprise, allowing for better decision-making. With TPS, you control the factors critical to production and essential to making profits: quality, yield, costs, cycle time, and schedules—while translating business information strategies into improved performance.

### Expanded control domain benefits

TPS provides a wide range of expanded control domain benefits, including:

- Improved safety
- Increased yield and throughput
- Minimized product “giveaway”
- Sustained plant capabilities and uptime
- Improved product consistency, flexibility, and production schedules
- Reduced waste, energy use, product cost, maintenance, and support.

**Figure 4 TPS unifies business and process control domains**



## 2. Interfacing to Control Systems

### 2.1 Introduction

#### About this section

This section describes several ways to interface to your control system. These interfaces include:

- Experion Station – TPS
- Experion Station – Console Extension (ES-CE)
- Experion Station – Flex (ES-F)
- Global User Station (GUS)
- Icon series console furniture
- Operator station displays
- SafeView
- *GUSPro*

This section also includes an overview of the Experion Server – TPS (ESVT). While not an operator console, it is a key component in the integration of TPS and Experion.

## 2.2 Experion Station – TPS

### About the Experion Station – TPS

Experion offers a replacement for the Global User Station (GUS) and Universal Station (US) functionality. Experion provides seamless integration with TPS. All TPS data, events, and operator messages are incorporated into Experion, such that TPS appears as an extension of Experion. Experion Station – TPS provides a common operator interface for both TPS and Experion. It directly connects to its local **TotalPlant** Network (TPN) through an LCNP board. It interfaces to other Experion systems through the Experion Server and Distributed System Architecture (DSA). These remote systems may include Experion-TPS systems. The direct connection to the TPN (also known as the LCN) is the basis of its seamless integration.

GUS components—Native Window, GUS Display Builder, GUS Display Runtime, and GUS*Pro* components—are supported on the Experion Station – TPS. The GUS Display Builder functionality is unchanged in the Experion Station – TPS. Access to HCI/OPC data (for example, data on a third party OPC server) from GUS displays continues to be possible using the GUS HCI Client.



## 2.3 Experion Station – Console Extension

### About the Experion Station – Console Extension

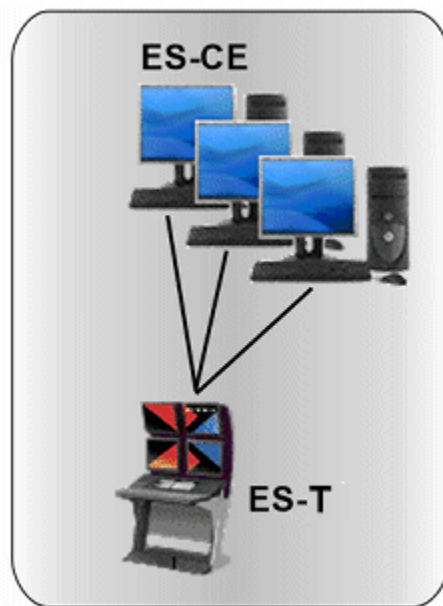
The Experion Station – Console Extension (ES-CE), when enabled with TPS options, is a Windows-based station running as an extended client to the host Experion Station – TPS. The ES-CE can access TPS data or Experion data through the Experion Station – TPS. An ES-CE station may reside in the Control Room or it may reside on an engineer's desktop. Possible uses for an ES-CE station include:

- Process monitoring
- Control and display definition using Control Builder and/or HMIWeb Display Builder
- Running other Control System applications for process analysis or maintenance using current data or history data

The ES-CE supports remote Native Window functionality.

The ES-CE with TPS options supports remote GUS Display functionality.

The ES-CE is also available in Icon Series console furniture.



## 2. Interfacing to Control Systems

### 2.4. Experion Station – Flex

---

## 2.4 Experion Station – Flex

### About the Experion Station – Flex

The Experion Station – Flex (ES-F) is a Windows-based station running as a client to the Server. It connects to the Server over the plant intranet—an Ethernet or Fault Tolerant Ethernet network. The ES-F can access either TPS data or Experion data through the Server.

An ES-F station may reside in the Control Room or it may reside on an engineer's desktop. Possible uses for an ES-F station include:

- Process monitoring
- Control and display definition using Control Builder and/or HMIWeb Display Builder
- Running other Control System applications for process analysis or maintenance using current data or history data

The ES-F supports remote Native Window functionality.

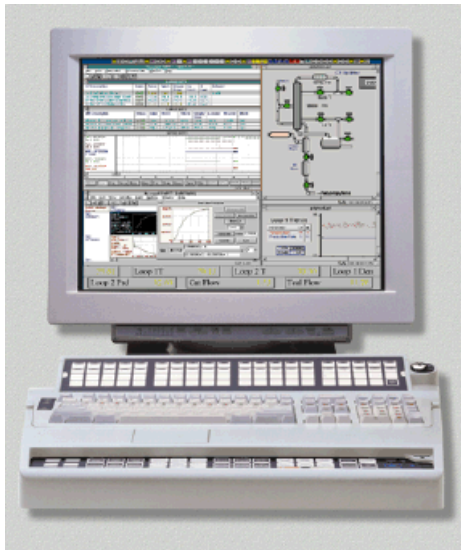
The ES-F is also available in Icon Series console furniture.



## 2.5 Global User Station

### About the Global User Station

The Global User Station (GUS) provides a graphical interface that helps you view multiple processes and events in real-time. GUS fully and seamlessly integrates the open Windows environment with a real-time control system. Its platform supports a suite of integrated functions for process operators and process engineers. GUS process operator functions connect operators to standard pre-configured operating, alarm, and system displays. As many as eight custom operating displays can be installed per GUS.



### The Role of GUS

GUS delivers multiple CRTs and process Windows per station, integrates process data and business information on every operator display, and uses ActiveX and OPC technologies. Its remote functionality provides operators with increased control capabilities for your plant.

This Windows XP station presents a wide variety of data without interfering with normal process control operations. Operators can focus on the main process display, while managing other windows that provide auxiliary process and system information.

## **2.6 Icon Series Console**

### **About the Icon Series console**

The Honeywell Icon Series console is an ergonomically designed console that promotes operator productivity. Its ease-of-use and comfort help optimize operator and engineering productivity. Unique Icon Console features include a modular, flexible design with an adjustable work surface panel, keyboard height, and screen angle.

**Figure 5 Honeywell Icon Series console**

The Icon Series Console delivers functionality in an ergonomically designed package.

#### **Features:**

- Pop-up disk drive; hot-swappable input and pointing devices.
- Adjustable work surface expands up to eight inches in height.
- Keyboard pop-up module is accessible and removable from front of console.
- Tilt/swivel dual or quad Flat Panel Displays (FPDs) help reduce glare and operator fatigue.
- Designed for compatibility with TPS and other Honeywell systems.  
Available in quad and dual flat panel display configurations with 18.1-inch and 20.1-inch screens.



## 2.7 Operator Station Displays

### Access to control and process information

Experion Station – TPS and GUS support displays that help operators react quickly and safely to changing situations. These displays:

- Simplify data access.
- Establish direct communication with applications.
- Support the Honeywell Communication Interface (HCI) and OPC standards.

The displays operate in a windows environment, allowing users to navigate among the process displays and other applications with the click of a mouse or a touch of the screen.

### Types of displays

Standard operating display categories include:

- **Plant process displays**—Overview, group, detail, and alarm summary displays allow control and monitoring of alarms, events, process trends, and numeric data.
- **TPS system displays**—System status associated with the status and performance of system hardware and software.
- **System function displays**—Point summaries provide access to system functions, including organizational summaries and reporting functions.

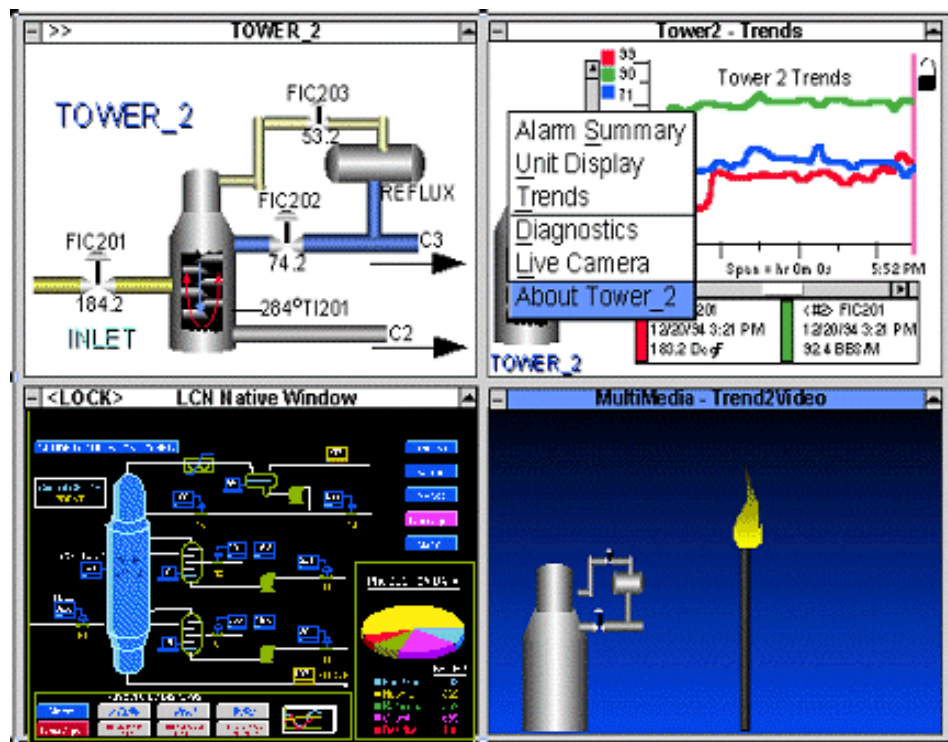
## 2.8 SafeView

### About SafeView

SafeView, the optional Honeywell Windows manager application, safeguards your plant, personnel, and assets by ensuring that critical operating displays are never obstructed by other displays or applications. With SafeView, engineers or operators can designate the screen location for each type of display—creating a repeatable environment tailored for various plant operations.

Operators can access and view data in multiple windows from different sources, giving them complete control over every window in their system. Existing applications automatically close when a new function is placed into a window.

**Figure 6 SafeView helps you to view critical plant operations**



## 2.9 GUSPro

### Applications for alarm management, event analysis and advanced trending

Twenty-five percent of operator-related incidents are attributable to operators receiving incorrect, missing, or confusing information. Honeywell developed the optional *GUSPro* productivity application to provide operators with applications for alert notifications, alarm management, event analysis and advanced trending.

### Using *GUSPro* tools

*GUSPro*'s integrated operating environment provides tools to help protect profitable operational performance at your plant. These tools include:

- ProMenu
- User Alert™
- Alarm Configuration Manager
- Event Analyst
- ProTrend™

### Developed in conformance with ASM practices

Developed under the guidance of the Honeywell-led Abnormal Situation Management (ASM) Consortium, *GUSPro* functionality is powered by practices proven to maximize effectiveness and sustain plant profitability.

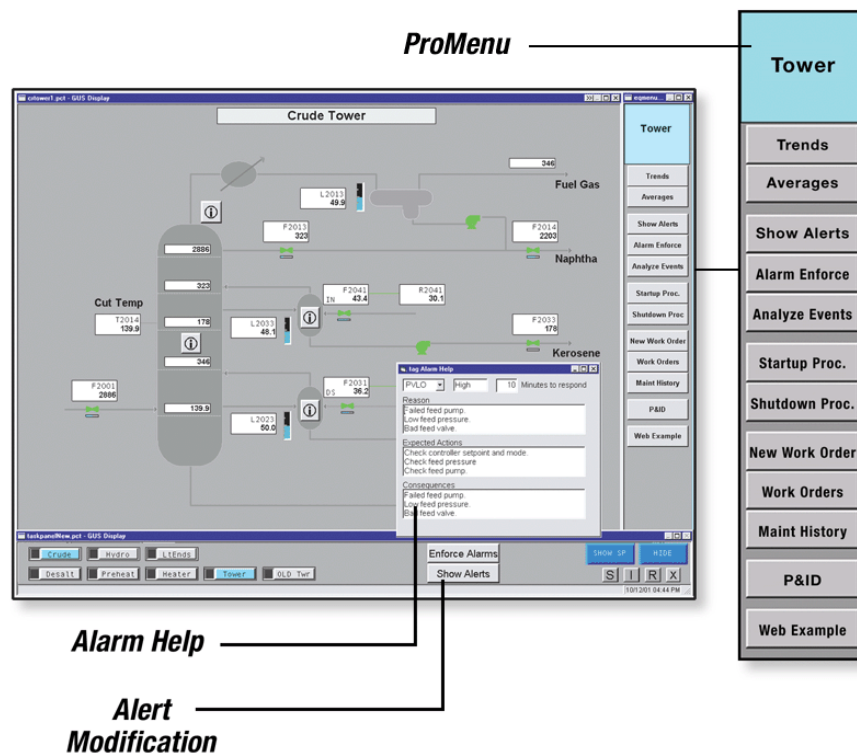
These ASM practices protect plant performance by ensuring operations are safe, environmentally responsible, and incident-free.

## 2. Interfacing to Control Systems

### 2.9. GUSPro

#### ProMenu

ProMenu provides operators with direct access to essential information from GUS process graphics. ProMenu provides rapid, direct, one-click access to all relevant information regarding a particular tag or process equipment. This includes alarm help, detail display, equipment maintenance information, operating procedures, historical trends, and loop diagrams.

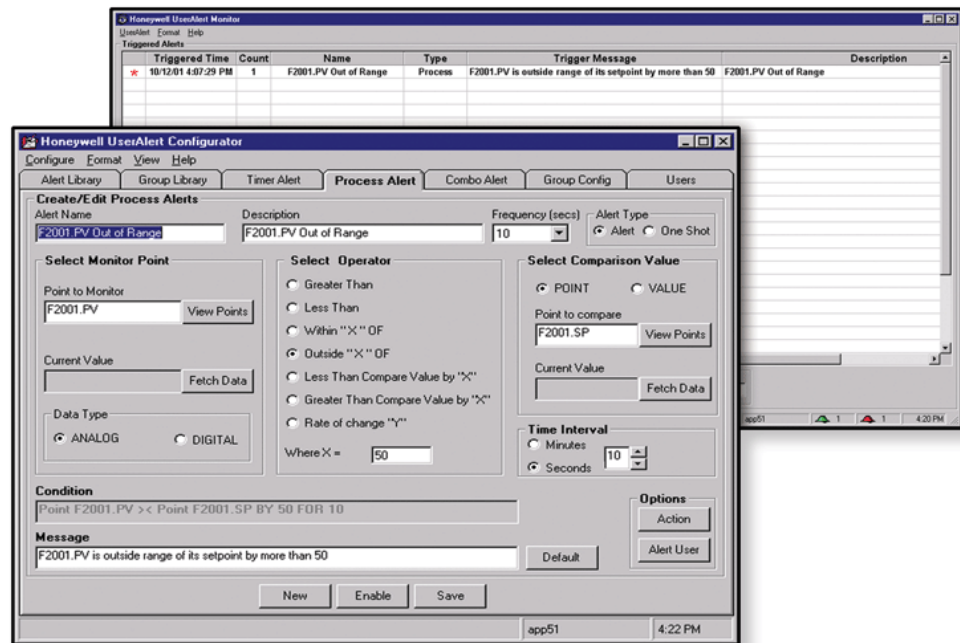


## UserAlert™

UserAlert, a configurable GUSPro application, automates alert monitoring and notification, and greatly reduces the operator's load.

UserAlert also allows non-essential notifications to be configured as alerts rather than alarms—this minimizes alarm flooding during an upset condition. UserAlert delivers notifications in the form of messages or audible alerts through the GUS, e-mail, or pager.

## UserAlert



## 2. Interfacing to Control Systems

### 2.9. GUSPro

---

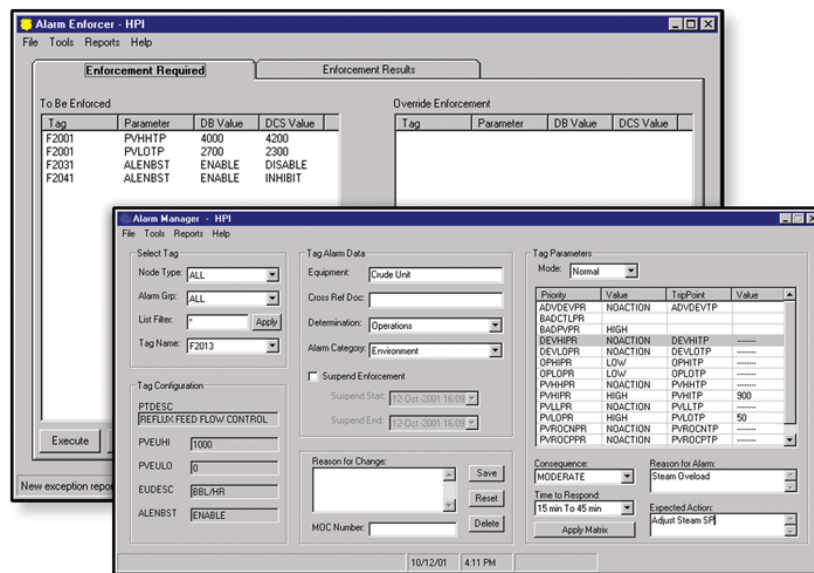
#### Alarm Configuration Manager

Alarm Configuration Manager helps you engineer, configure, and manage changes precisely to match the requirements of your plant's alarm system.

GUSPro's Alarm Configuration Manager extends the capability of traditional alarm systems by capturing expertise and knowledge of your best operator and engineer for integration within the alarm system. The alarm information (including the reason for the alarm, probable consequences, required time to respond, and recommended action) is available to the operator as alarm help, directly from the operating display.

Alarm Configuration Manager supports the definition of multiple alarm settings for each plant or equipment operating state. Monitoring and optional enforcement can be scheduled or performed manually as needed.

#### *Alarm Configuration Manager*



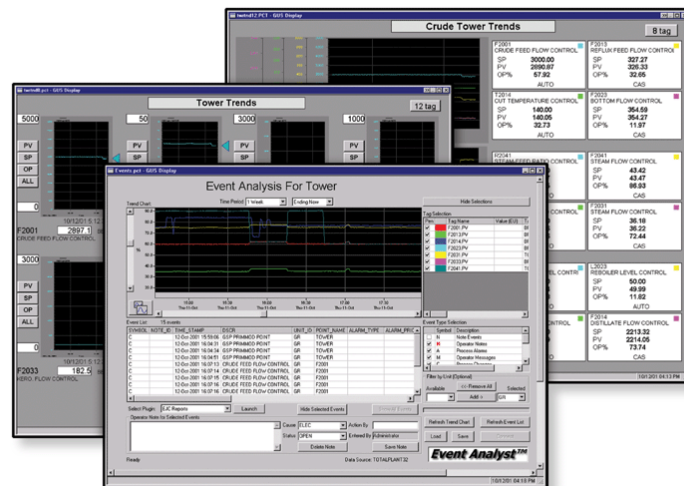
## Event Analyst and ProTrend

Event Analyst analyzes the alarm log, history database, and process change journal for abnormal situations within a process unit. Event Analyst performs these tasks:

- Allows correlation between event data and process history data, aiding in the identification of cause and effect relationships.
- Helps users visualize previously unseen and undetected process relationships, selected process variables, and all types of TPS events, including alarm journal events, message journal events, and operator changes.

The ProTrend application provides advanced trending tools to maximize operator awareness and response to potential, or actual, abnormal situations. Operational trends provide operators with process history context when making process changes with predefined trend sets and real-time operational data. Using operator-defined trends, operators can add tags to trends dynamically from a display by clicking on a tag or piece of equipment. Using Event Analyst and ProTrend, operators can detect and analyze significant events. This information is vital for detection of, and recovery from, process upsets.

### *Event Analyst and ProTrend*



## **2.10 Experion Server – TPS**

### **About the Experion Server – TPS**

The Experion Server – TPS provides all of the functionality of the Experion Server, plus additional hardware and software components specific to TPS. Some of the features and functions of the Server are:

- Process and event history
- Available in redundant configuration
- LCNP for direct connection to the TPN
- Host services for operator stations (clients)
- Repository services for Experion components
- OPC Server for data and events for advanced control and optimization applications
- Distributed System Architecture (DSA) that enables seamless integration of multiple Experion servers. This integration includes data, alarms, history, trends, and reports.

## 3. Controlling the Process

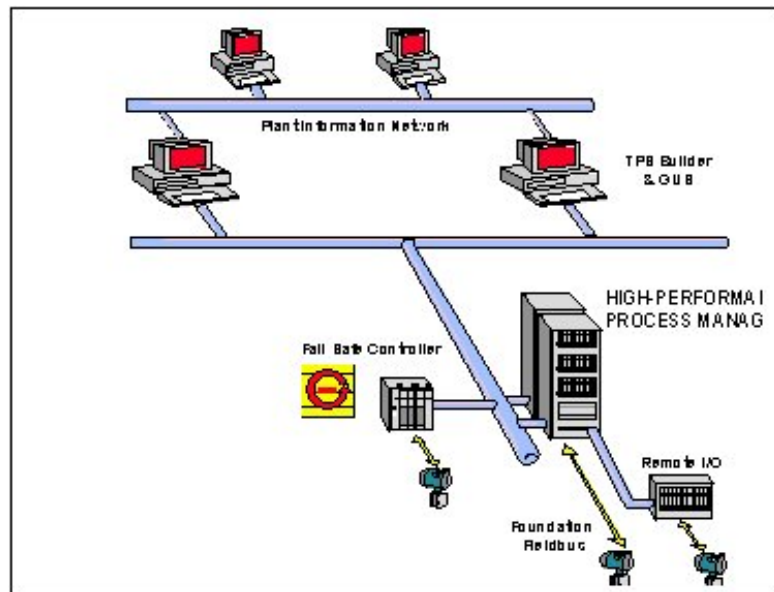
### 3.1 Honeywell Process Control Products

#### Helping you to control your world

This section describes Honeywell process control products which include:

- High-Performance Process Manager (HPM)
- Fails Safe Controller (FSC)
- Field Instruments
- TPS Field I/O
- TPS System Tools
- Experion controllers

**Figure 7 Honeywell controllers and field instruments integration**



## 3.2 High-Performance Process Manager

### About the High-Performance Process Manager

The High-Performance Process Manager (HPM) is a premier control platform, especially for large, critical processes. It provides the following features:

- A reliable, redundant-configuration.
- Deterministic control ensures consistent operation of control applications.
- Multi-processor architecture has separate microprocessors to perform specific tasks.

With an extensive set of algorithms, the HPM is the solution for continuous and discrete logic applications. The HPM integrates third-party devices directly into the controller, providing a complete, single-source solution. By using Control Language in conjunction with the standard algorithm set, you can implement custom applications for solving control problems for batch, continuous, sequencing, and hybrid applications.

### HPM architecture

The HPM's architecture consists of:

- A 32-bit architecture
- Parallel-processing using multiple processors
- Each I/O module contains its own microprocessor
- Advanced, secure process control memory system
- Separate processors for control and communications
- Every I/O Processor individually manages its own memory requirements with on-board RAM

## 3.3 Fail Safe Controller

### Help safeguard your process operations

Honeywell Safety Management Solutions (HSMS) has been the industry leader in the field of process safeguarding systems for a quarter of a century. The integration of the Fail Safe Controller (FSC) with the TPS system can enhance the safety of processes, plants, and personnel. All FSC configurations are tightly integrated with the TPS system, providing flexibility in designing, configuring, monitoring, controlling, and safeguarding your safety-critical applications.



### Fail Safe Controller

The innovative FSC design ensures certainty and reliability of operation. The FSC system is a user-programmable, Dual-Modular Redundant (DMR) or Quadruple-Modular Redundant (QMR®) microprocessor-based safety system that performs a wide range of high-integrity process control and safety functions, including:

- High-integrity process control
- Burner/boiler management system
- Process safeguarding and emergency shutdown
- Turbine and compressor safeguarding
- Fire and gas detection systems
- Pipeline monitoring

## **3.4 Honeywell Field Instruments**

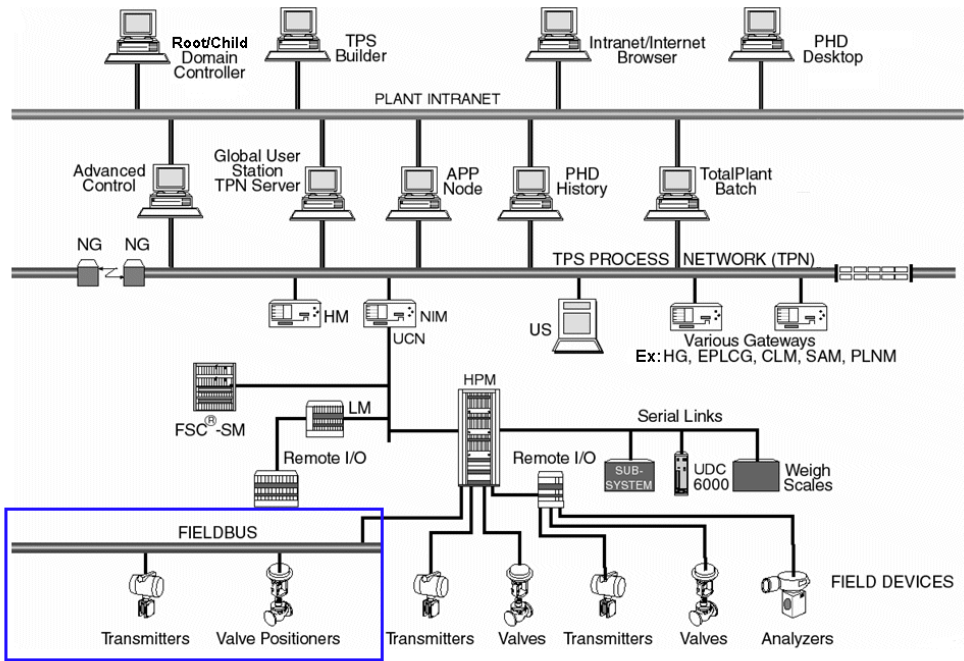
### **Accurate, reliable measurement**

Honeywell provides integrated instrumentation products for temperature, pressure, level, flow, and specialty measurements.

### **Supporting FOUNDATION Fieldbus**

Honeywell's field instruments support FOUNDATION Fieldbus (FF) with registered FF devices, including the complete ST 3000 family of products (such as differential, gauge, absolute, level, and remote seal) and the STT35F Temperature Transmitter. Control is secure all the way to the valve with Flowserve's Valtek Logix 1400 Fieldbus valve positioners, which also share Honeywell's implementation of FF. Tested and validated as a part of the total system, Honeywell and Valtek field devices are integral elements of Honeywell's control system architecture.

### Figure 8 Typical Honeywell Fieldbus configuration

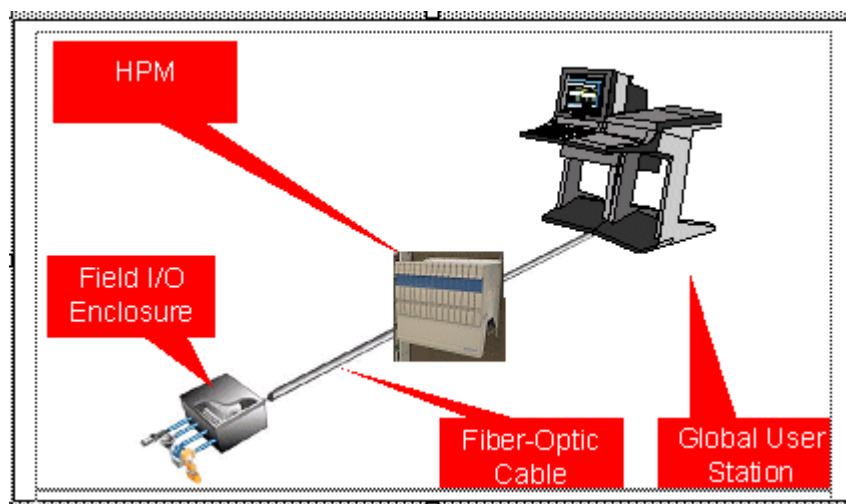


## **3.5 TPS Field I/O**

### **Field devices for extreme environments**

TPS Field I/O is a portfolio of remote I/O modules that can be located close to the process—eliminating control room-mounted I/O, marshalling cabinets, and multi-conductor field wiring. TPS Field I/O allows a group of any type of field devices can be connected to a single network node. Fully integrated with the TPS architecture, and field-mountable in a range of environmentally hardened enclosures, TPS Field I/O is engineered to perform under extreme temperatures and operate within corrosive and explosive gas environments.

**Figure 9 Field I/O configuration**



TPS Field I/O modules connect over a fiber-optic link to the HPM, the TPS system's robust controller. Featuring undetectable integration, the HPM recognizes TPS Field I/O points as if they were local. This solution supports as many as 40 remote locations for each HPM, while offering compact I/O packaging that supports eight to 256 channels for each remote location.

## 3.6 TPS System Tools

These tools integrate with TPS system, helping unify disparate systems.

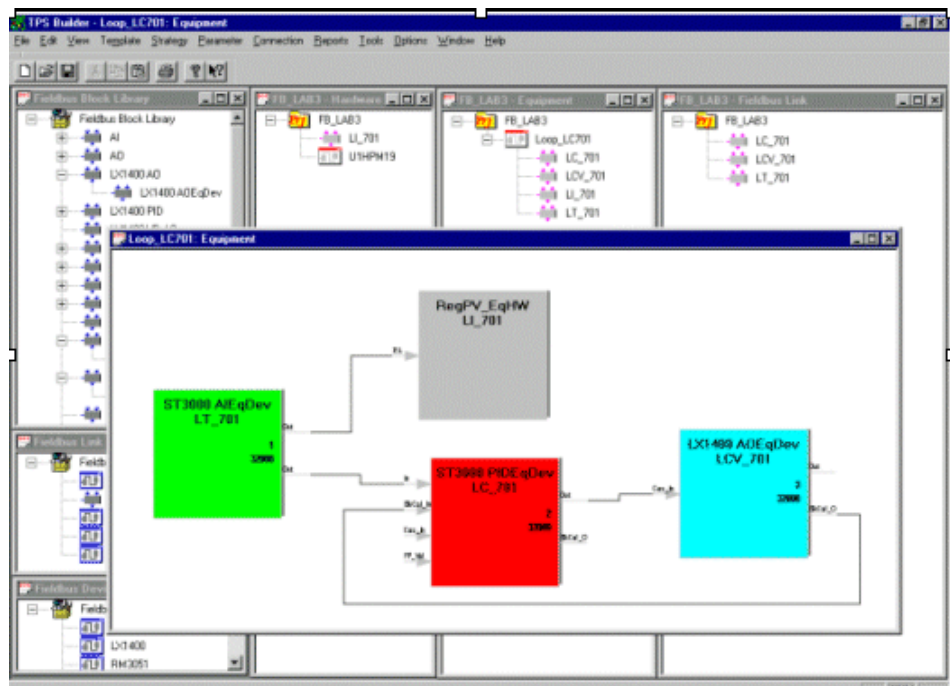
**Picture Builder**—integrates a wide variety of data to increase the effectiveness of the process operator.

**DOC3000 database management tools**—enhances engineering productivity, and reduces costs associated with documenting and maintaining control systems.

**Additional engineering tools**—enables implementation of complex control loops and control strategies.

**TPS Builder, Global User Station (GUS) Display Builder, and HMIWeb Display Builder**—helps you build and configure control strategies on a TotalPlant Solution system.

Figure 10 TPS Builder helps you to build and configure control strategies



### **3. Controlling the Process**

#### **3.6. TPS System Tools**

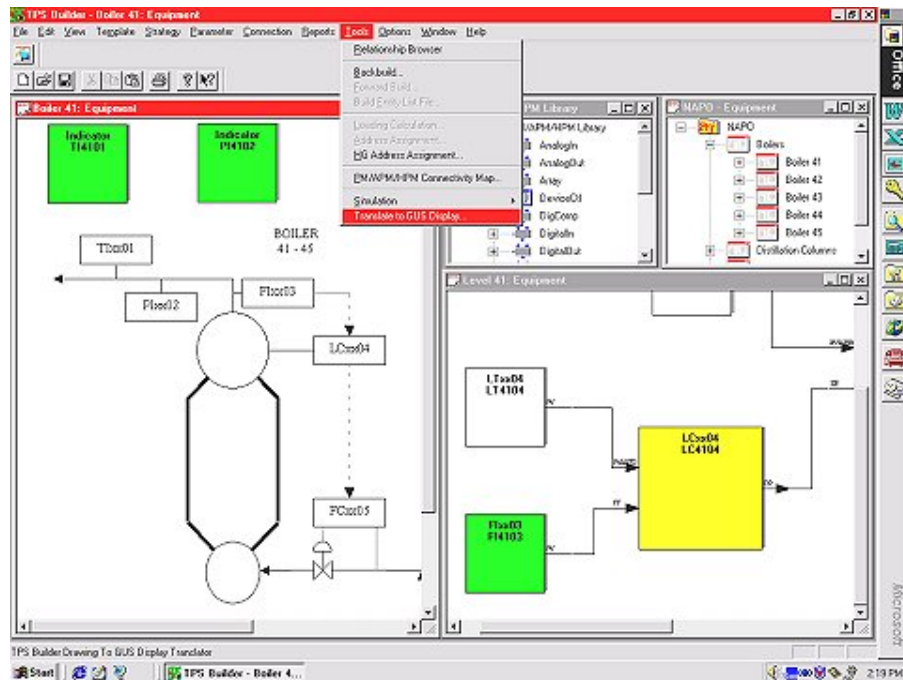
---

##### **TPS Builder**

TPS Builder is a graphical engineering software tool for building control strategies and configuring process control data on a TPS system. TPS Builder includes:

- Database reporting
- Control strategy drawing
- Tools for building and using templates
- An easy-to-use graphical user-interface
- Control Language (CL) programming support
- Ability to share data and work with other applications
- The ability to simultaneously create a control drawing and the control strategy

Figure 11 TPS Builder tools for control strategy display



### 3. Controlling the Process

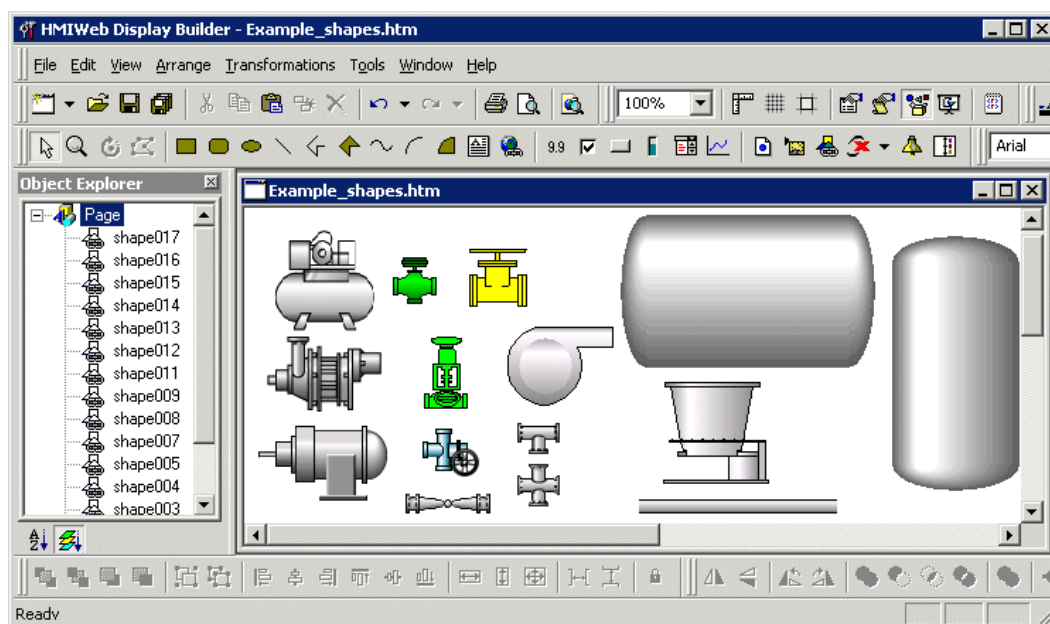
#### 3.6. TPS System Tools

---

#### HMIWeb Display Builder

You can create custom displays that represent your process using HMIWeb Display Builder. This tool runs on Experion stations, including Experion Station – TPS (ES-T). HMIWeb Display Builder is based on Internet technology. Displays are saved in HTML and are displayed by a browser. HMIWeb Display Builder provides a gallery of shapes to facilitate display building. The following example shows just a few of the many pre-built shapes.

**Figure 12 HMIWeb Display Builder gallery**



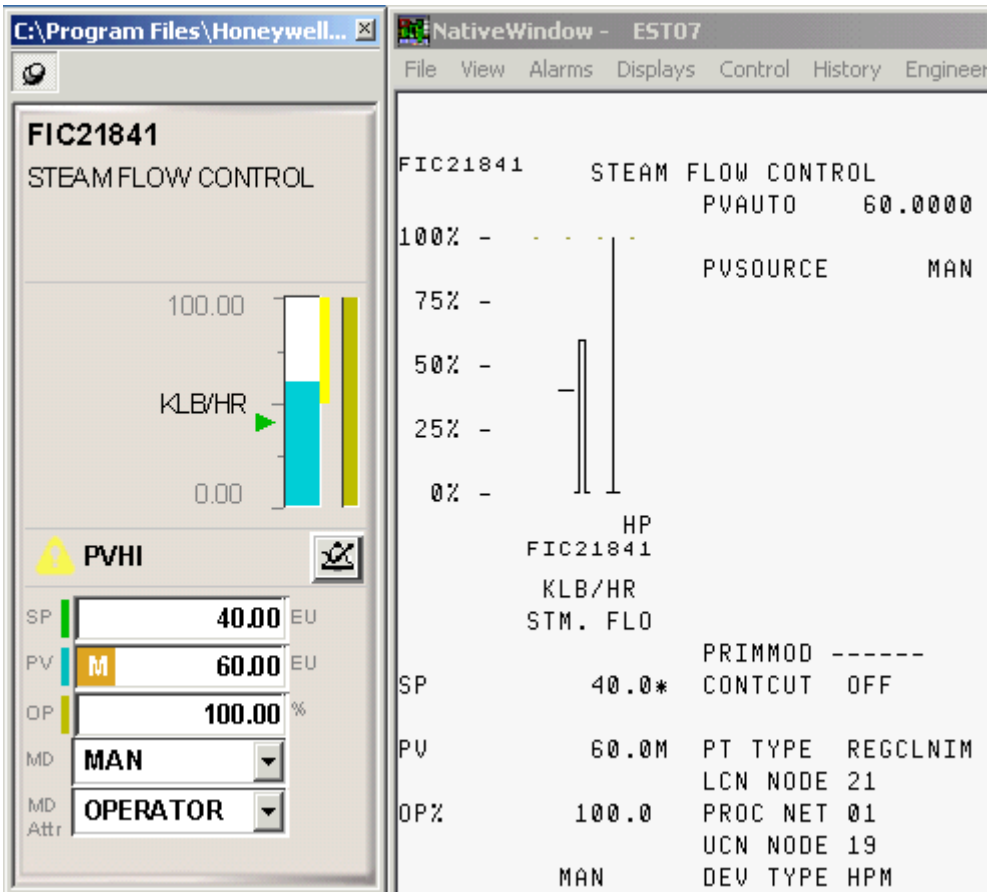
#### REFERENCE - INTERNAL

For more information about HMIWeb Display Building, refer to the *HMIWeb Display Building Guide*.

HMIWeb Faceplates for TPN points

The HMIWeb-based faceplate displays support the TPN’s AM, NIM, HG point types. The HMIWeb-based faceplate provides a replacement for the classic TPN faceplate that is also supported in Experion. The following example shows an example HMIWeb faceplate displaying a Native Window’s TPN control point.

Figure 13 TPN Point in HMIWeb faceplate



## **3.7 Accessing Experion Controllers**

### **Accessing Experion controllers from TPS**

Integrating TPS with Experion allows you to access Experion process controllers that are located on a remote Experion platform. Some of the components and technologies that make this integration possible are:

- Experion Station – TPS (ES-T)
- Experion Station – Flex (ES-F)
- Experion Server – TPS (ESVT)
- Distributed System Architecture (DSA)

A TPS system that has LCN-connected Experion nodes can communicate directly with Experion control as well as connect with other (remote) Experion systems by means of DSA. DSA allows consoles and applications on the TPS system to access the Experion controllers located on the remote Experion systems.

### Direct Control Integration

Both the Server and Console Station, while LCN-connected, can also directly connect to Experion controllers as well as support ACE. The Experion controllers use FTE connections. The Experion controllers are not directly connected to the TPN.



### LCN-connected ACE

Application Control Environment (ACE) acts as a supervisory controller in which Experion control strategies execute. When an ACE becomes LCN-connected, the platform for the ACE is the standard ACE platform, with the addition of a LCNP card and selected components of the APP version of the TPS infrastructure.

An LCN-connected ACE provides all ACE functions, with no extensions. An LCN-connected ACE has the same relationship with TPN data owners as an ACE. On an Experion system, a LCN-connected ACE can participate in peer-to-peer relationships with other Experion controllers. On a TPN system, a LCN-connected ACE functions as a HCI Data Access client of TPN data owners, using its local TPN Server and LCNP.

### **3. Controlling the Process**

#### **3.7. Accessing Experion Controllers**

---

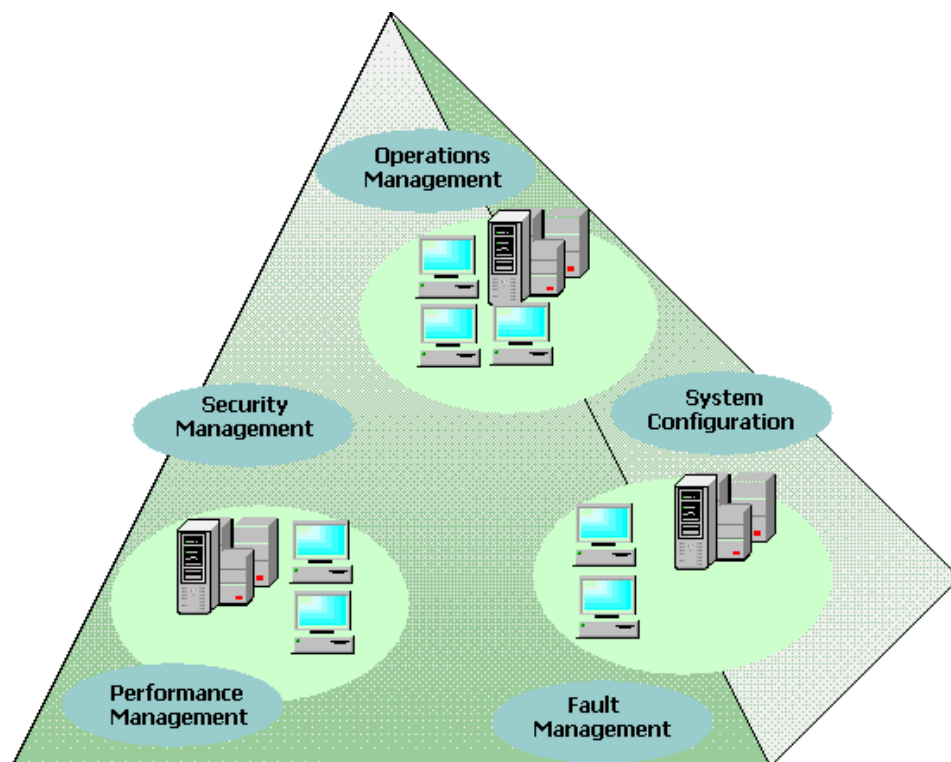
## 4. Managing Systems and Networks

### 4.1 System Management

#### About System Management

System Management includes system configuration, operations management, performance management, fault management and security management. Figure 14 shows the various functions of System Management.

Figure 14 Role of system management



#### What does System Management provide?

## **4. Managing Systems and Networks**

### **4.1. System Management**

---

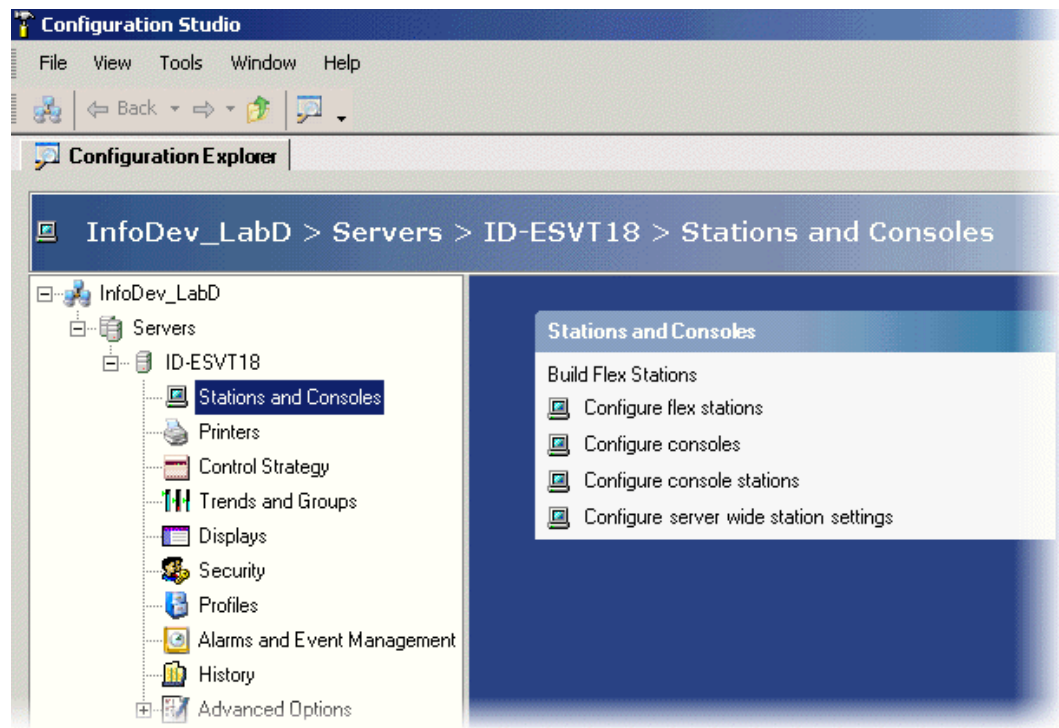
Honeywell System Management software provides the following tools:

- HCI name service
- Status of managed nodes
- Remote TPS standard configuration
- Remote HCI component configuration
- Monitoring of local HCI components
- Monitoring of local Redirection Managers
- Management of local HCI components
- System Management Display based within a Microsoft Management Console (MMC)
- Windows System Events integration (workstation-logged events in a TPS Domain Event log)

### Experion configuration tools

Experion engineering tools are available from the Experion Configuration Studio. Configuration Studio itself provides a common configuration environment for hosting and integrating configuration tools. It represents a single, common desktop application for the integration and/or invocation of the Experion tool set.

**Figure 15 Experion Configuration Studio display**



## 4.2 System Management Operations

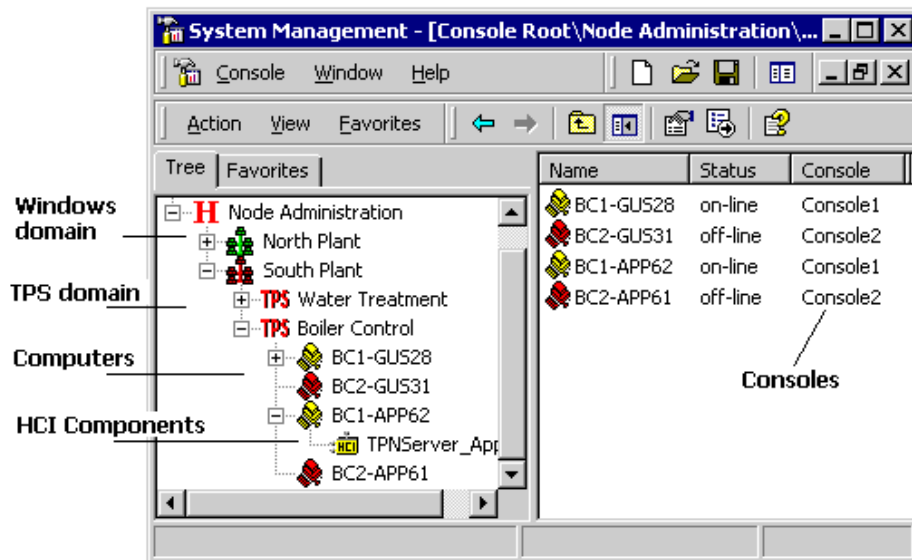
### Introduction to System Management

System Management is the management of your process control system using Honeywell tools. Most system management tasks can be accomplished through the System Management display.

### What is the System Management display?

The System Management Display is your main user interface for System Management. The System Management Display provides a hierarchical view of the Windows Domain, TPS Domain, Computer, and Managed Component Status. When TPS Domains are organized into consoles, the consoles appear in the results pane of the display. The System Management Display appears within a Microsoft Management Console (MMC), as shown in the following figure.

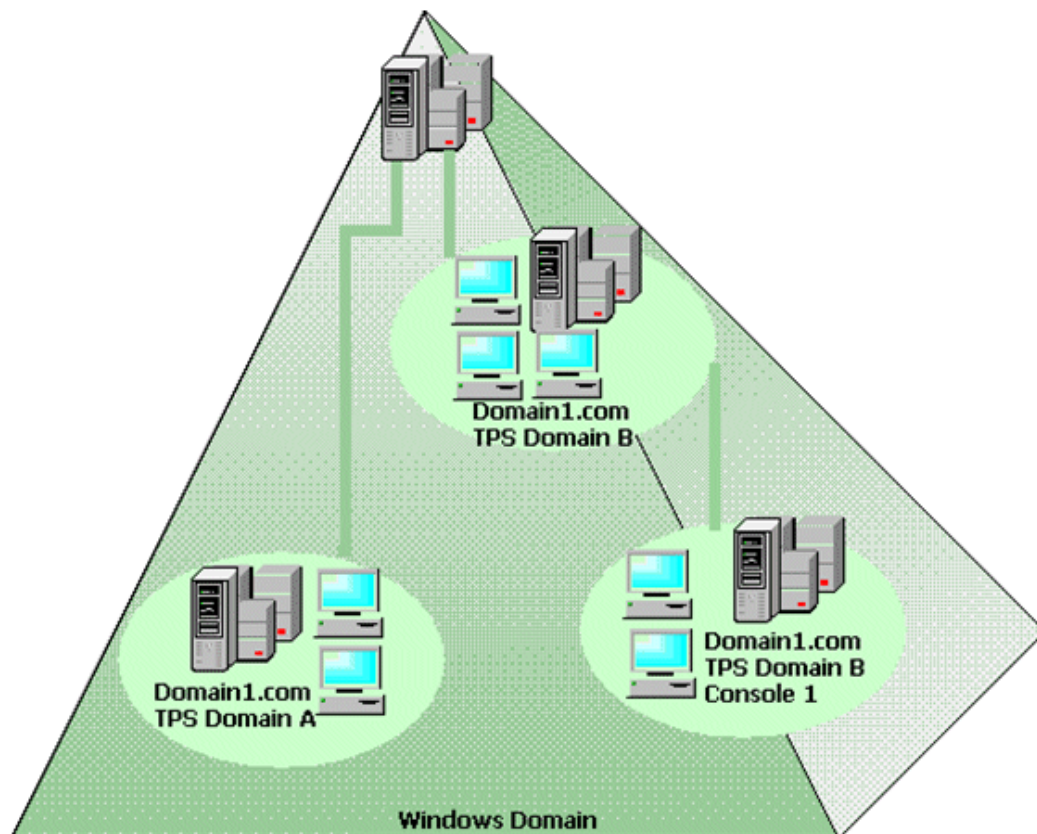
**Figure 16 Typical System Management display**



## 4.3 TPS Domains and Consoles

A TPS Domain or a Console is derived from an Organizational Unit in the Windows 2003 Active Directory. By creating a TPS Domain or console, you can organize and manage plant nodes from a System Management Display. Windows 2003 Plant Control Network nodes can be organized along the lines of classic DCS systems, into areas of interest or responsibility.

Figure 17 TPS Domains in a Windows domain



## **4. Managing Systems and Networks**

### **4.3. TPS Domains and Consoles**

---

#### **Using Windows 2003 system configuration tools**

Use TPS Builder to define or configure process control strategies.

Windows 2003 provides tools for configuring a TPS system and includes the following:

- TPS nodes
- Time synchronization
- Attributes of HCI-managed components
- HCI-managed components (such as HCI TPN Server)
- Performance parameters and policies (such as throttling)
- TPS Domain as an Organizational Unit within Active Directory

## 5. Supporting Control Communications

### 5.1 Open System Technologies and Products

#### Introduction

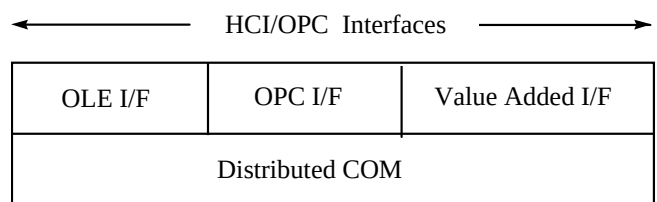
Honeywell uses industry-standard products and technologies that provide secure communication between major TPS components, and allows these components to be physically distributed across TPS nodes.

#### Open System technologies

These include:

- **COM/DCOM**—Component Object Modeling/Distributed COM
- **OPC**—OLE for Process Control
- **HCI**—Honeywell Communication Interface
- **Windows**—Microsoft Operating System

**Figure 18 Architecture of HCI/OPC communication infrastructure**



17399

#### TPS system communications products and technologies

TPS system communications products and technologies include:

- Fault Tolerant Ethernet (FTE)
- Redirection Manager (RDM)
- Distributed System Architecture (DSA)

## **5.2 COM/DCOM and OPC**

### **About system support**

The TPS system fully supports OLE to allow embedded objects in displays directly linked to applications (such as spreadsheets, safety documents, and emergency instructions) that are within the system and on PCs linked to the plant Intranet. This feature provides operators access to business and control information on the same display, resulting in efficient communication among applications and system components.

### **COM/DCOM**

Distributed Component Object Model (DCOM) technology provides a communication infrastructure that allows Honeywell-supplied TPS system components and third-party applications to act as clients to TPS system data sources. Third-party data sources that conform to the OPC standard can also be data servers to TPS system components. DCOM is designed for use across multiple network transports, including Internet protocols such as HTTP. DCOM is based on the Open Software Foundation's DCE-RPC specification and will work with both Java applets and ActiveX components through the Component Object Model (COM).

### **DCOM works with HCI/OPC**

Honeywell Communications Interfaces (HCI) use DCOM technology and OPC. HCI includes data access mechanisms defined by the OPC standards committee and enhancements, such as prioritized requests, timed requests, and status information. The Honeywell Communication Interface provides:

- Safety attributes.
- Critical reliability.
- Improved ease of use.
- Improved data access.
- Increased uptime for advanced applications.
- Decreased application-development and testing time.

## **5.3 HCI/OPC**

### **About HCI/OPC**

HCI extensions are used by the Application Processing Platform (APP) node to provide applications access to real-time control data while protecting control security.

HCI/OPC has three main parts:

1. The minimum DCOM interfaces that all servers must implement.
2. The OPC standard interfaces that OPC-compliant servers must implement.
3. The value-added interfaces that Honeywell defines are implemented in one or more of the HCI/OPC servers, including management interfaces required for HCI-managed components.

## **5.4 Fault Tolerant Ethernet**

### **About Fault Tolerant Ethernet**

Honeywell's Fault Tolerant Ethernet (FTE) provides a way to achieve Ethernet redundancy through the use of Honeywell's FTE driver in conjunction with redundant, commercially available LAN hardware. Using FTE, network communications take place over a functioning route. If that route should fail, communication continues over the alternate route. With FTE, a system can recover from single faults and, in some cases, from multiple faults.

A fault tolerant Ethernet increases system reliability, availability, and plant productivity. FTE plays an important part in the supervisory and control portions of the process-connected layers of the TotalPlant Solution system.

### **Using FTE with new and existing systems**

The FTE redundancy system is optional and layered over standard hardware, operating systems, and application level software. The redundancy hardware and software can be optionally applied to previously shipped TPS and PlantScape® systems, as well as to new systems, such as Experion PKS.

## 5. Supporting Control Communications

### 5.4. Fault Tolerant Ethernet

---

#### How does FTE work?

An FTE network uses redundant switches and cables. There are four communication paths between any two FTE nodes. An FTE network connects clusters of nodes. A cluster is a group of nodes with high intercommunication, typically associated with the same process unit.

FTE is an option for:

- TPS GUS, APP, PHD
- Experion Servers and Stations
- Switch: Up to 96 ports in 12-port increments
- PMD Server, Station, Field Controller (required)
- Other applications on certified PCs with Windows 2003/XP
- Firewall: Required between FTE network and any other network
- FTE network: Up to 511 FTE nodes (dual-connected) and up to 511 standard nodes (singly-connected)

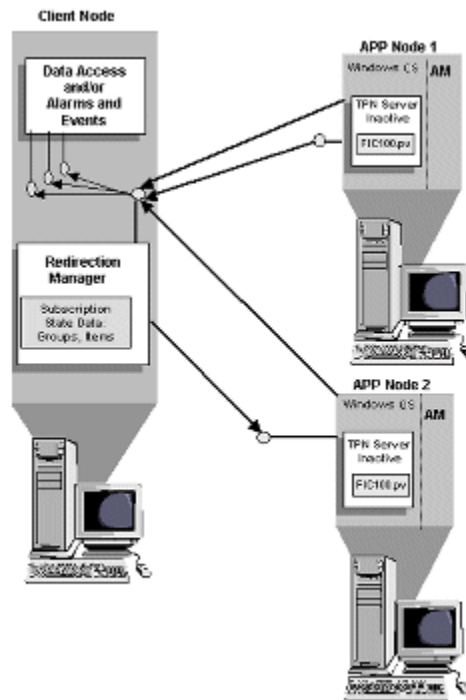
## 5.5 Redirection Manager

### About Redirection Manager

The Redirection Manager (RDM) is a software option that enables clients to transparently switch to a secondary server if the primary server fails. RDM can be used to implement redundant TPN Servers for high availability. Redirection Manager provides your TPS system with high availability/reliability. After redirection completes, the client application continues to perform its OPC-related tasks as usual.

Figure 19 shows how an optimization application can access data from TPS APP Node through Redirection Manager. You can configure the optimization application to use APP Node 1 as its primary data source and APP Node 2 as its secondary data source.

**Figure 19 How Redirection Manager works**



## 5.6 Distributed System Architecture

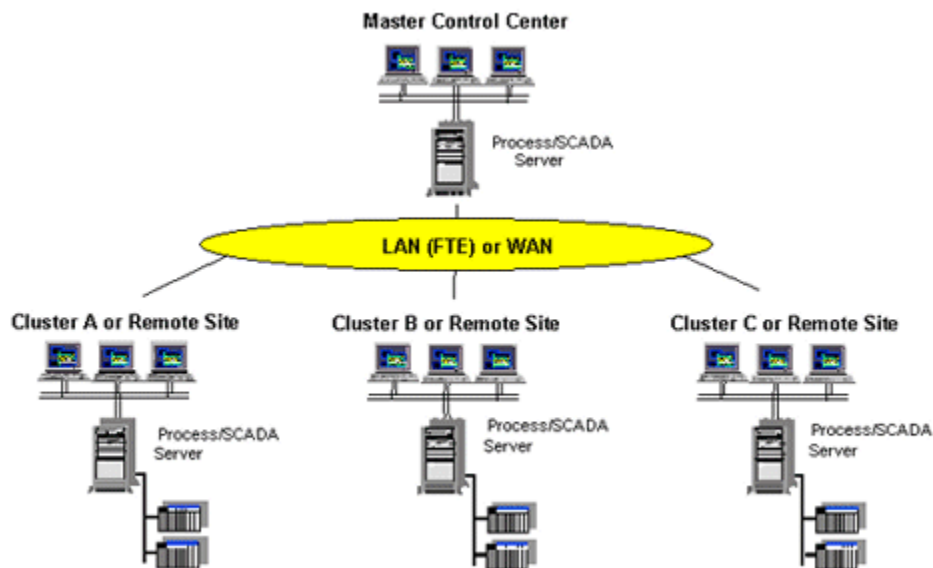
### About Distributed System Architecture

Distributed System Architecture (DSA) enables you to integrate multiple Experion servers (including Experion Server – TPS) into a single operational system. DSA allows these servers to share data, alarms, messages, and history without the need for duplicate configuration on any server. DSA is effective for:

- Geographically distributed systems.
- Integrated systems containing both TPS and Experion controllers.
- Logically separate Experion systems located in different parts of a plant.
- Combinations of the above.

The following figure shows a geographically distributed system in which DSA would be effective.

**Figure 20 Geographically distributed system**



## 6. Integrating with Control and Business Needs

### 6.1 Application Processing Platform

#### About the Application Processing Platform

The Application Processing Platform (APP) is the TPS system's advanced applications platform. It supports the APP node, PHD node, Integration Node, and CLM node. The APP combines the security and integrity needed for advanced process control and optimization. The APP can run complex applications developed by Honeywell, by users, and by third-party suppliers. Integrating process and business data, the APP uses industry-standard technologies, such as the Microsoft Windows 2003 operating system, and communication interfaces, such as DCOM, OLE, and OPC.

**Figure 21 Application Processing Platform (APP)**



## **6.2 Control and Business Process Integration**

### **PHD software integrates business and control information.**

Process History Database (PHD) software works seamlessly with the TPS system's control domain, bringing data from all other sources into its integrated structure to serve information needs plant-wide—across multiple control systems and environments. Its integration of business and control information makes implementation and maintenance of applications easier, faster, and less costly.

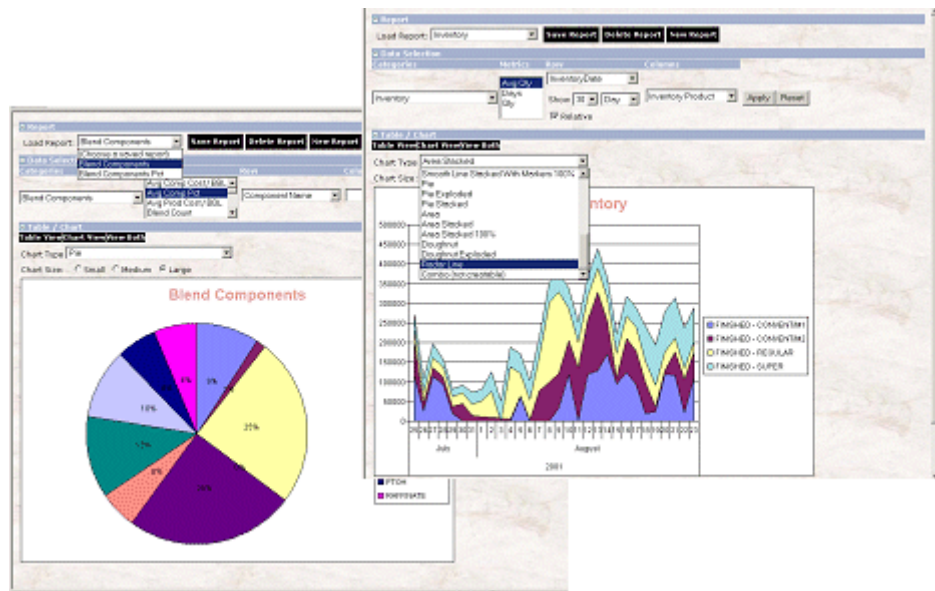
PHD establishes a single, open, integrated architecture for plant-wide data collection, storage, and management. A common database system supports and integrates all plant and business data, as well as all applications and procedures. Its common database system gives plant personnel easy desktop access to timely and accurate information. PHD software combines knowledge of your plant with process and other types of data, thus providing information, not just data, to applications and users. The result is that your plant operators make informed decisions, instead of chasing data and relying on guesswork.

## 6.3 Business.FLEX: Improved Decision Making and Profitability

Business.FLEX solutions help customers improve plant profitability by enabling plant staff to work more effectively and make better decisions. The solutions support collaborative manufacturing by offering increased visibility into plant operations and across the supply chain.

Business.FLEX solutions build on both PHD and the TPS system to enable the most comprehensive approach available for active plant and enterprise management. Business.FLEX solutions allow enterprise objectives to be effectively executed at the manufacturing level, converting business objectives into operational targets and returning validated production data—effectively closing the loop on the business planning cycle. With this capability, you can typically increase production by 2 to 5 percent and decrease costs by 0.5 to 1 percent.

Figure 22 Business.FLEX supports collaborative manufacturing



## **6.4 Services to Protect Your Technology Investment**

Honeywell is committed to protecting your intellectual and physical capital, as well as helping you optimize the results received from your process control technology investment. Life Cycle Management (LCM) Services feature a complete portfolio of solutions that extend the performance of your TPS system by providing a path to the latest technology. LCM Services also help maximize the capabilities of your Honeywell hardware and software applications.

### **Solution Enhancement Support Services**

The Honeywell Solution Enhancement Support Program (SESP) for the TPS system is a component of Life Cycle Management (LCM) Services. SESP provides complete life-cycle management capability—including continuation, refresh, and evolution services. This flexible support program, co-developed by Honeywell and our customers, provides flexibility to choose the option that best matches your site's specific needs. From around-the-clock telephone technical assistance, software upgrades, hardware upgrade kits, and enhancements, to specialized quarterly reports and preferred customer pricing, there is an SESP program designed to meet the needs of your plant.

## **6.5 Consistent Evolution**

The Honeywell promise of Consistent Evolution provides options that integrate your installed system with newer Honeywell technologies. Honeywell has made this commitment to our customers—which comprise the largest installed based in the industry—for more than 30 years. Whether you need long-term support of your existing system, or you want to augment your TPS system with the latest technologies, interoperate your TPS system with a new system, or migrate to a new system based on existing capital and intellectual investments, you can be assured that your investment in Honeywell technology is protected.



**Honeywell**

---

Honeywell  
Process Solutions  
2500 West Union Hills Drive  
Phoenix, AZ 85027 USA